

3.2 Classwork: Trigonometry

1. As shown, right $\triangle ABC$ has $AC = 8$, $BC = 15$, $AB = 17$, $m\angle C = 90^\circ$.

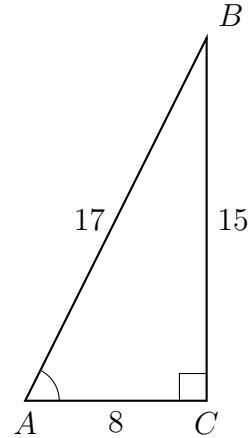
Express each trigonometric ratio as a fraction.

(a) $\sin A =$

(b) $\cos A =$

(c) $\tan A =$

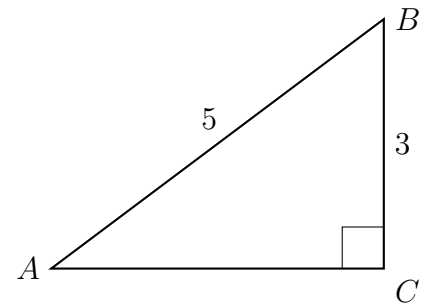
- (d) Find $m\angle A$, to the *nearest degree*.



2. Right triangle $\triangle ABC$ is shown with measures as marked.

- (a) Write down $\sin A$.

- (b) Find the length of side AC .



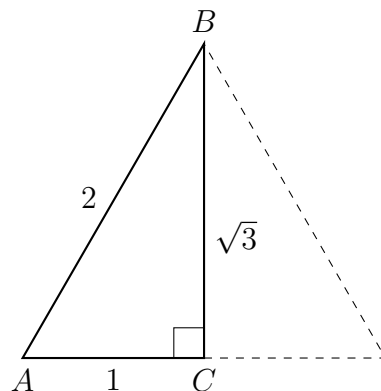
- (c) Find the angle measure of $\angle A$, rounded to the *nearest degree*.

3. Right $\triangle ABC$ has base $AC = 1$, height $BC = \sqrt{3}$, and hypotenuse $AB = 2$ as marked. (A reflection $\triangle ABC$ of is also shown.)

(a) Write down the angle measure of $\angle A$.

(b) Write down the angle measure of $\angle ABC$.

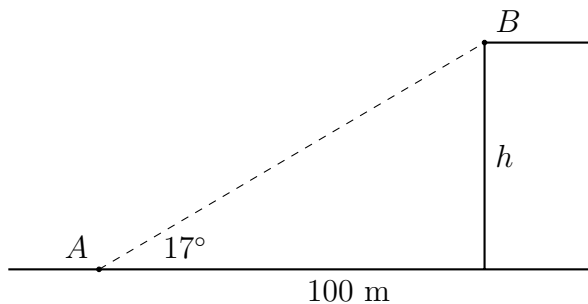
(c) Write down $\cos A$.



4. At an angle of elevation of 17° , the top of a structure B is visible from point A on the ground 100 meters away, as shown below.

Find the height h of the structure to the *nearest meter*.

(not to scale)



5. A pirate is looking down from the top of a mast with a height of 12 meters. Below him, the pirate sees an enemy ship 100 meters away.

Find the angle of depression to the *nearest degree*.

